

Volumes from Known Cross Sections and Cylindrical Shells

Answer Key

AP Calculus AB/BC

Quick Answers

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|---|---|
| 1. the volume = $\frac{64}{3} \text{ m}^3$ | 5. the volume = $8\pi \text{ m}^3$ |
| 2. the volume = $\frac{81\pi}{16} \text{ cm}^3$ | 6. the volume = $\frac{972\pi}{5} \text{ cm}^3$, — |
| 3. the volume = $\frac{81\pi}{2} \text{ ft}^3$ | 7. the x-values where the curves meet = 0, 1, |
| 4. the volume = $\frac{16\sqrt{3}}{3} \text{ cm}^3$ | the volume = $\frac{1}{30} \text{ m}^3$ |
| | 8. the volume = $\frac{40\pi}{3} \text{ cm}^3$, — |
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What is verified

9 of 11 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 11 tagged checks.

1. Base: region bounded by $y = x$, the x -axis and $x = 4$ (metres). Cross sections perpendicular to the x -axis are squares. Find the volume.

Solution: At position x the side of the square runs from the x -axis up to the line, so $s = x$ and $A(x) = x^2$ square metres. Slicing perpendicular to the x -axis,

$$V = \int_0^4 x^2 dx = \left[\frac{x^3}{3} \right]_0^4 = \frac{64}{3}.$$

So the volume is

$V = \frac{64}{3} \approx 21.33 \text{ m}^3$

2. Base: region bounded by $y = \sqrt{x}$, the x -axis and $x = 9$ (centimetres). Cross sections perpendicular to the x -axis are semicircles on the base. Find the volume.

Solution: The diameter is $d = \sqrt{x}$, so the radius is $\frac{\sqrt{x}}{2}$ and

$$A(x) = \frac{1}{2}\pi \left(\frac{\sqrt{x}}{2} \right)^2 = \frac{\pi x}{8}.$$

Then $V = \int_0^9 \frac{\pi x}{8} dx = \frac{\pi}{8} \left[\frac{x^2}{2} \right]_0^9 = \frac{81\pi}{16}$.

$$V = \frac{81\pi}{16} \approx 15.90 \text{ cm}^3$$

3. Region bounded by $y = x^2$, the x -axis and $x = 3$ (feet), revolved about the y -axis. Use shells.

Solution: A shell at position x has radius x and height $y = x^2$, so

$$V = \int_0^3 2\pi x \cdot x^2 dx = \int_0^3 2\pi x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^3 = \frac{81\pi}{2}.$$

So the volume is

$$V = \frac{81\pi}{2} \approx 127.23 \text{ ft}^3$$

4. Base: triangle bounded by $y = 4 - x$ and the axes (centimetres). Cross sections perpendicular to the x -axis are equilateral triangles. Find the volume.

Solution: The side at position x is $s = 4 - x$, so $A(x) = \frac{\sqrt{3}}{4}(4 - x)^2$ and the region runs from $x = 0$ to $x = 4$:

$$V = \int_0^4 \frac{\sqrt{3}}{4}(4 - x)^2 dx = \frac{\sqrt{3}}{4} \left[-\frac{(4 - x)^3}{3} \right]_0^4 = \frac{\sqrt{3}}{4} \cdot \frac{64}{3} = \frac{16\sqrt{3}}{3}.$$

So the volume is

$$V = \frac{16\sqrt{3}}{3} \approx 9.24 \text{ cm}^3$$

5. Region bounded by $y = 4 - x^2$ and the x -axis, $x \geq 0$ (metres), revolved about the y -axis. Use shells.

Solution: The region runs from $x = 0$ to $x = 2$ (where $4 - x^2 = 0$). A shell at x has radius x and height $4 - x^2$:

$$V = \int_0^2 2\pi x (4 - x^2) dx = 2\pi \left[2x^2 - \frac{x^4}{4} \right]_0^2 = 2\pi(8 - 4) = 8\pi.$$

So the volume is

$$V = 8\pi \approx 25.13 \text{ m}^3$$

6. Region bounded by $y = \sqrt{x}$, the x -axis and $x = 9$ (centimetres), revolved about the y -axis. (a) Volume by shells. (b) Why does the shell setup use the curve as written?

Solution (a): Shell radius x , shell height $\sqrt{x}$:

$$V = \int_0^9 2\pi x \sqrt{x} dx = \int_0^9 2\pi x^{3/2} dx = 2\pi \left[\frac{2}{5} x^{5/2} \right]_0^9 = \frac{4\pi}{5} (243) = \frac{972\pi}{5}.$$

So the volume is

$$V = \frac{972\pi}{5} \approx 610.73 \text{ cm}^3$$

Solution (b) — instructor-judged. Shells about the y -axis integrate in x , so $y = \sqrt{x}$ can be substituted for the shell height exactly as given. Washers about the y -axis integrate in y , which forces the curve to be rewritten as $x = y^2$ and splits the solid into an outer cylinder

of radius 9 minus the region left of $x = y^2$. Full credit names the variable of integration as the reason.

7. Base: region enclosed by $y = x$ and $y = x^2$ (metres); square cross sections perpendicular to the x -axis. (a) Where do the curves meet? (b) Find the volume.

Solution (a): Set $x = x^2$, i.e. $x - x^2 = x(1 - x) = 0$.

$$x = 0, x = 1$$

Solution (b): On $[0, 1]$ the line is above the parabola, so the side of the square is $s = x - x^2$ and

$$V = \int_0^1 (x - x^2)^2 dx = \int_0^1 (x^2 - 2x^3 + x^4) dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \frac{1}{30}.$$

So the volume is

$$V = \frac{1}{30} \approx 0.03 \text{ m}^3$$

8. Region bounded by $y = x^2$, the x -axis and $x = 2$ (centimetres), revolved about the line $x = 4$. (a) Volume by shells. (b) Why is the radius $4 - x$?

Solution (a): The shell at position x sits a distance $4 - x$ from the axis and has height x^2 :

$$V = \int_0^2 2\pi(4 - x)x^2 dx = 2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^2 = 2\pi \left(\frac{32}{3} - 4 \right) = \frac{40\pi}{3}.$$

So the volume is

$$V = \frac{40\pi}{3} \approx 41.89 \text{ cm}^3$$

Solution (b) — instructor-judged. The shell radius is the distance from the axis of revolution to the shell, and the axis $x = 4$ lies to the right of the whole region, so that distance is $4 - x$. Checking the ends: at $x = 0$ the shell is 4 units away, at $x = 2$ it is 2 units away — the radius shrinks as the shell approaches the axis, which is what the picture shows. Using x would measure from the y -axis instead, which is not the axis of revolution here.